Secure Remote Network Access Using Twin Gate and Raspberry Pi

Business Requirements Document

Date

Revision History

Date	Version	Author	Change

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Business Requirements Document defines the general business requirements for the project. Also identifies business and end user requirements, problems or issues, project information, process information, and training and documentation requirements.

1 Purpose

The purpose of this project is to develop a secure, efficient, scalable, and cost-effective remote network access solution using a Raspberry Pi as a connector for the Twingate Zero Trust Network Access (ZTNA) platform.

- Traditional VPNs require complex configurations such as firewall changes, port forwarding, and public IP addresses, while exposing internal services to security risks.
- Provide secure, controlled, and cost-effective remote access using Zero Trust principles, identity verification, and least-privilege access.
- Configure a Raspberry Pi as a Twingate connector and integrate the Twingate API with Python automation for secure, scalable, and centrally managed network access.

2 Project Information

This project aims to provide secure and cost-effective remote network access using a Raspberry Pi with Twingate ZTNA. It focuses on improving security, simplifying network access, and reducing manual configuration. The system supports access control, user authentication, API-based automation, DNS aliasing, temporary access, and activity logging.

2.1 Project Description

This project provides a secure and cost-effective remote network access solution using a Raspberry Pi as a Twingate ZTNA connector. It enables authorized users to securely access internal network resources without complex VPN configurations, port forwarding, or public IP addresses. The system uses Zero Trust principles, access control, authentication, and activity logging, with Python-based Twingate API automation to simplify resource management and improve scalability.

2.2 Project Approach

The project will be developed through a series of planned phases to ensure secure, reliable, and efficient remote network access. Each phase contributes to the overall goal of implementing a Raspberry Pi-based Twingate Zero Trust Network Access (ZTNA) solution.

Phase 1: Requirement Analysis
The first phase involves identifying the requirements of the project, including network resources, user access needs, security requirements, and system limitations. The requirements will be analyzed to determine how Twingate and the Raspberry Pi can be used to provide secure remote access.

Phase 2: System Design
In this phase, the overall system architecture will be designed. The Raspberry Pi will be planned as the Twingate connector between the internal network and the Twingate cloud platform. User authentication, access policies, network resources, and communication flow will also be defined.

Phase 3: Raspberry Pi and Twingate Setup
The Raspberry Pi will be configured and connected to the internal network. Twingate Connector software will be installed and configured on the Raspberry Pi. The Twingate platform will then be configured to establish secure communication between authorized remote users and internal network resources.

Phase 4: Security and Access Control
Security policies will be implemented based on Zero Trust principles. User authentication, device verification, and least-privilege access will be configured so that users can access only the resources they are authorized to use. Activity logging will also be enabled for monitoring and accountability.

Phase 5: API Automation
A Python-based automation script will be developed using the Twingate API. The script will help discover and manage network resources dynamically, reducing manual configuration. Additional features such as DNS aliasing, temporary access permissions, and service accounts can also be integrated where required.

Phase 6: Testing and Validation
The complete system will be tested to verify connectivity, authentication, access control, security policies, and API automation. Unauthorized access attempts will also be tested to ensure that restricted resources remain protected.

Phase 7: Deployment and Documentation
After successful testing, the system will be deployed for intended use. Configuration details, system architecture, security policies, automation procedures, and testing results will be documented. This final phase ensures that the solution is properly maintained and can be managed in the future..

2.3 Goals, Objectives, and Scope

The main goal of the project is to develop a secure, reliable, and cost-effective remote network access system using Raspberry Pi and Twingate ZTNA. The objectives are to:

- Provide secure remote access to internal network resources.
- Implement Zero Trust-based authentication and access control.
- Reduce the complexity of traditional VPN configuration.
- Automate network resource management using Python and the Twingate API.
- Enable monitoring and activity logging.

ID	Description	Included/ Excluded
01	Provide secure remote access to authorized internal network resources.	Included
02	Configure Raspberry Pi as a network connector for the Twingate platform.	Included
03	Implement Zero Trust-based authentication and access control.	Included
04	Apply least-privilege access so users can access only authorized resources.	Included
05	Reduce the complexity associated with traditional VPN configuration.	Included
06	Automate network resource discovery and registration using Python and the Twingate API.	Included
07	Support DNS aliasing for network resources.	Included

2.4 Business Drivers

The project “Secure Remote Network Access Using Twingate and Raspberry Pi” is being implemented to provide a secure, efficient, and cost-effective method for accessing internal network resources remotely**.**

- Reduce infrastructure and deployment costs**:** Use a Raspberry Pi as a lightweight and affordable network connector instead of requiring complex dedicated network infrastructure.
- Improve security**:** Implement Zero Trust Network Access (ZTNA) principles to ensure that users are authenticated and can access only the resources for which they are authorized.
- Simplify remote access**:** Reduce the complexity associated with traditional VPN-based remote access, including firewall modifications, port forwarding, and public IP requirements.

2.5 Stakeholders

The stakeholders for the Secure Remote Network Access Using Twingate and Raspberry Pi project are the individuals and groups who are involved in the development, supervision, implementation, usage, and evaluation of the proposed solution**.**

Name	Department	Role
Akshay N	Computer Science & Engineering (Cybersecurity)	Project Developer / Student
Abhishek k b	Computer Science & Engineering (Cybersecurity)	Project Developer / Student
Shanid Ali	Computer Science & Engineering (Cybersecurity)	Project Developer / Student
Ms. Farbeena N	Computer Science & Engineering	Project Guide

2.6 Assumptions, Dependencies, and Constraints

This section identifies the assumptions, dependencies, and constraints that may affect the development, testing, deployment, and operation of the Secure Remote Network Access Using Twingate and Raspberry Pi project.

2.6.1 Assumptions

- A Raspberry Pi is available for configuration as the Twingate network connector.
- The required internal network resources are available for testing.
- Authorized users have valid credentials and appropriate permissions.
- Users have reliable internet connectivity for remote access.
- The required Twingate services are available during development and testing.

2.6.2 Dependencies

- Raspberry Pi – Required to act as the network connector.
- Twingate ZTNA Platform – Required to provide secure remote access, authentication, and access control.
- Twingate API – Required for automated network resource discovery and registration.
- Python – Required for implementing the automation component.
- Network Connectivity – Required for communication between remote users, Twingate, the Raspberry Pi connector, and internal resources.

2.6.3 Constraints

- The performance of the solution may depend on the processing and network capabilities of the Raspberry Pi.
- Remote access depends on the availability of internet connectivity.
- Access to internal resources is restricted according to configured authorization and least-privilege policies.

- Dependence on Twingate services and API availability may affect system operation and automation.
- Hardware and resource availability may limit the scale of implementation and testing.

2.7 Risks

The following risks may affect the development, testing, deployment, and operation of the Secure Remote Network Access Using Twingate and Raspberry Pi project.

Risk 1: Raspberry Pi Hardware Failure
Failure of the Raspberry Pi may interrupt the secure remote access service.
Mitigation: Maintain a backup configuration and restore the system or replace the Raspberry Pi when required

Risk 2: Internet Connectivity Failure
Loss of internet connectivity may prevent remote users from accessing internal network resources.
Mitigation: Ensure a stable internet connection and use an alternative connection where available.

Risk 3: Incorrect Access Configuration
Incorrect access policies may allow unauthorized access or prevent authorized users from accessing required resources.
Mitigation: Apply least-privilege access policies and thoroughly test user permissions before deployment

Risk 4: Authentication Failure
Authentication problems may prevent authorized users from accessing network resources.
Mitigation: Verify authentication settings and ensure appropriate account recovery procedures are available.

Risk 5: Twingate API or Automation Failure
Failure of the Twingate API or Python automation script may prevent automatic network resource discovery and registration.
Mitigation: Test the automation component thoroughly and maintain a manual configuration procedure as a backup

Risk 6: Unauthorized Access
Improper configuration or compromised credentials could result in unauthorized access to internal resources.
Mitigation: Use identity verification, device verification, authentication, and least-privilege access controls.

2.8 Costs

Provide estimated total costs for implementing the proposed solution. These amounts should be detailed in a Project Capital and Expense Costs Worksheet, if applicable. Note any assumptions used to estimate the costs.

2.9 Delivery Dates

List high-level deliverables / milestones for the project and their associated target dates.

Milestones / Deliverables	Target Date

3 Process Information

3.1 Current Processes

Provide Business Process Diagrams for any complex processes or steps that are used to do a specific job. Consider systems involved, users, when tasks are performed, and what the results are. Describe the process using text or a graphic process flow.

Any business rules, such as calculations, decisions / if-then, algorithms, or procedures should be clearly defined and broken down to their individual steps. Include the following information in the process diagrams:

- ***Inputs, processes, and outputs***
- ***Business workflow***
- ***Business rules, edits, validations, and/or formulas***
- ***Exception handling and processing.***

3.1.1 Current Process Flow

Provide detailed information about current processes.

3.1.2 Current Process Description

Provide detailed step information about current processes in the following table, if applicable.

#	Description	User	Issues
1			
2			
3			
4			
5			

3.2 New Processes or Future Enhancements

Provide Business Process Diagrams for any new complex processes or steps that must be performed to do a specific job. Consider systems involved, users, when tasks are performed, and what the results are. Describe the process using text or a graphic process flow.

Any business rules, such as calculations, decisions / if-then, algorithms, or procedures should be clearly defined and broken down to the individual steps. Include the following information in the process diagrams:

- *Inputs, processes, and outputs*
- *Business workflow*
- *Business rules, edits, validations, and/or formulas*
- *Exception handling and processing.*

3.2.1 New Process Flow

Provide detailed information about the new processes.

3.2.2 New Process Description

Provide detailed step information about the new processes in the following table, if applicable.

#	Description	User	Issues
1			
2			
3			
4			
5			

4 Requirements Information

4.1 High-Level Business Requirements

This section includes details about the business requirements. Business requirements are those items needed to support user goals, tasks, and activities. It may include high-level modifications or enhancements. They can be either mandatory or desirable.

- ***Mandatory: "Must Have" represents the core functionality and must be included in the solution.***
- ***Desirable: "Nice to Have" requirements can be implemented after all mandatory requirements are fulfilled and there are sufficient resources available.***

This section should also include business rules that impact the project. A Business Rule defines or constrains some aspect of the business, e.g., products sold with a 50% or more discount must include an authorized approval code.

#	Function	Description	Priority	Comments

Note: Priority may be defined as Mandatory or Desirable.

4.2 System Interfaces

Provide information about the systems or applications that the solution must interface with, e.g.,

- ***Communication hardware, software and their requirements, methods, and functionality.***
- ***Data, formats, messages, and transfer schedules.***
- ***Performance and capacity.***
- ***Security designs and considerations.***
- ***Names of reference manuals and other documentation along with their location.***

4.3 Infrastructure Requirements

Provide infrastructure information when there are equipment / network additions or changes, e.g., servers, printers, network devices, network configuration or management changes, new or upgraded middleware or operating system software, or changes to data centers.

5 Other Requirements Information

5.1 Product, Organizational, System, and External Requirements

This section includes details about other requirements that can cover products, organizations, systems (e.g., performance, operations), and external requirements (e.g., security).

Audience	***Indicate the users of the system or application***
Security	***Provide information about the standard or unique security system or application situations, e.g., access / authentication / authorization processes; client, user or server certificates; encryption, password rules, and security procedures.***
Volume and Performance Metrics	***Indicate system or application volume and performance metrics to be performed, e.g., database records or transactions to be processed within a certain time period.***
Capacity and Scalability	***Indicate capacity and growth information or concerns that will have to be met over time.***
Support Considerations	***Indicate system or application support, e.g., who will support it, hardware / software to support it, service level agreements or contractual obligations.***
Audit	***Indicate any system or application audit requirements or constraints.***
Design Constraints	***Indicate system or application design constraints, e.g., software languages, developmental tools, architectural and design constraints.***
Output	***Indicate reports, files or other export output.***
Data Retention	***Indicate any special data retention requirements.***
Legal or Regulatory	***Indicate any legal or regulatory requirements that can impact the system or application.***

5.2 Usability, Performance, Operations, and Maintenance Requirements

Provide requirements information related to product usability, system performance, operations, maintenance or conditions that must be met by the proposed application or system. These requirements can have a more detailed title / description or be in a numbered list.

#	Function	Description	Priority	Comments

5.3 Content / Data / Sample Report Requirements

Include information about reports or data that must be provided or modified, e.g.,

- ***Samples of all content and data.***
- ***Examples of input / output files and their schema.***
- ***List all necessary content with a brief description.***
- ***Who will provide the information? List will provide the information and who is responsible for what and when, e.g., one-time only, other content must be maintained on a regular daily, weekly, or monthly basis.***
- ***Who will maintain the information when the application is operational?***

5.4 Screen Requirements

Provide any business rules or information related to screen information that must be included.

Element	Description & Rules	Source	Req	Default Value

5.5 Training and Documentation Requirements

Provide training and documentation information that is required to support the application or system to be implemented, e.g., training plans, training materials, support documentation, and user documentation.

Training / Documentation	Resource	Schedule	Comments

6 Glossary

List any document terms that may not be fully understood without some explanation.

Term	Definition
Business Requirements	Business requirements identify a necessary attribute, capability, characteristic, or quality of a system that has value and utility to a user. Sets of requirements are used as input into the design stages of product development. The requirements phase can be preceded by a feasibility study or a conceptual analysis phase of the project. Requirements-gather requirements from stakeholders. Analysis-check for consistency and completeness. Definitions-write down descriptive requirements for developers. Specifications-create details that are an initial bridge between requirements and design.
Functional Requirements	Functional requirements define the internal workings of the software, i.e., the calculations, technical details, data manipulations, and other specific functionality that show how the events are to be satisfied. The core of the requirement is the description of the required behavior, which must be a clear and readable description of the behavior. This behavior can come from organizational or business rules, or it can be discovered through working sessions with users, stakeholders, and other experts within the organization. Functional requirements generally contain a unique name and number, a brief summary, and reason for it. This information is used to help the reader understand why the requirement is needed, and to track the requirement through the development of the system. Functional requirements are supported by non-functional requirements, which impose constraints on the design or implementation.

Non-Functional Requirements	Non-functional requirements specify criteria that can be used to judge the operation of a system, rather than specific behaviors. They impose constraints on the design or implementation. Typical non-functional requirements are performance, reliability, security, scalability, and cost. Other terms for non-functional requirements are "constraints", "quality attributes" and "quality of service requirements."

7 APPENDIX